

Generation Interconnection System Impact Study Report

For Queue Project Q20-1514

Lone Tree 138 kV

July 28, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1514. The Interconnection Customer proposes a 139.7 MW solar photovoltaic facility interconnecting at the Lone Tree 138 kV substation in the Virginia area. The request was received on July 05, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 376 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,697,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CRAVEN COUNTY WOOD ENERGY LP-LOWMOOR 138 KV	OSCEOLA-LONE TREE 138 KV line	127.2	433	100.0
CRAVEN COUNTY WOOD ENERGY LP-LONE TREE 138 KV	LONE TREE SUBSTATION-OSCEOLA 138 KV	126.8	438	100.0
VC PORTERDALE HYDROELECTRIC-NEELY 138 KV Line	CRAVEN COUNTY WOOD ENERGY LP SUBSTATION-LONETREE 138 KV	100.4	338	42.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LOWMOOR-CRAVEN COUNTY WOOD ENERGY LP 138 KV Line	Rebuild line with higher-capacity conductor	\$11,424,000	25% (\$2,873,000)
LONE TREE-CRAVEN COUNTY WOOD ENERGY LP 138 KV	Rebuild line with higher-capacity conductor	\$10,712,000	25% (\$2,694,000)
VC PORTERDALE HYDROELECTRIC-NEELY 138 KV	Reconductor line	\$860,000	15% (\$130,000)
	Total allocated to project		\$5,697,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.